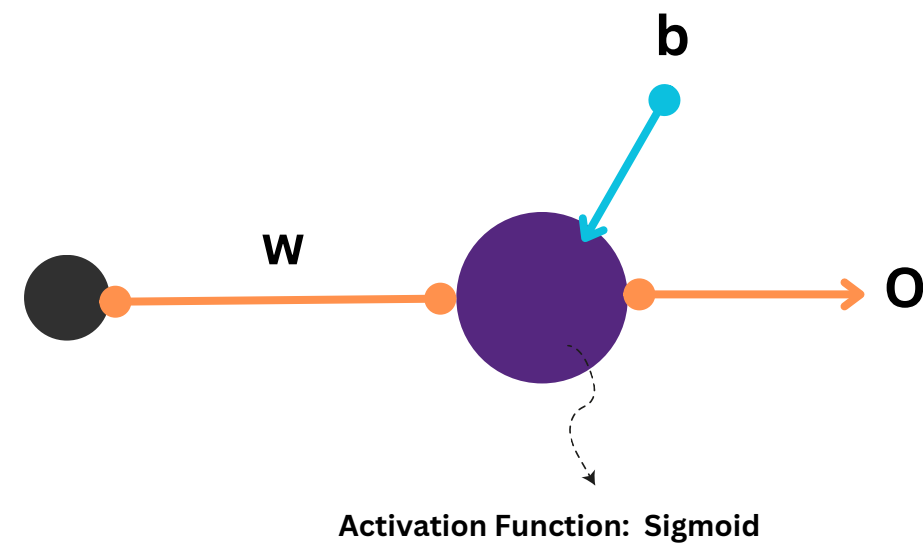




1. Cluster the **breast-cancer.csv** dataset with **KMeans** and compare it with the accuracy of **KNN**!

2. Obtain new value of weight and bias after Gradient descent first iteration, use MSE as a loss function.

x	y
1.3	2.7
2.1	4.5
7.6	14.9



Initial weight value(weight have been initialized randomly) : 1.5

Initial bias value (weight have been initialized randomly) : 0.5

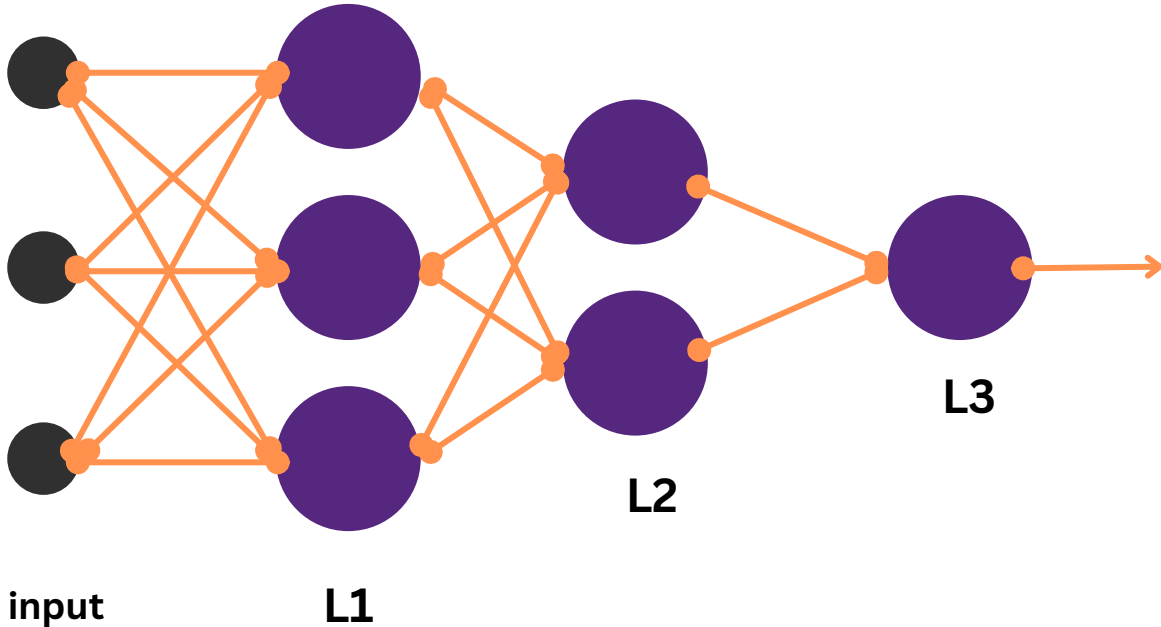
learning rate (alpha) = 0.1

3. For the below feed-forward neural network architecture obtain the following variables:

$O_1^1 = ?$

$O_2^2 = ?$

$O_1^3 = y_{\text{predicted}} = ?$



inputs

0.3
0.5
0.8

L1-Weights

0.1	0.2	0.2
0.4	0.1	0.2
0.2	0.3	0.1

L1-Biases

0.2
0.2
0.1

L2-Weights

0.1	0.2	0.2
0.4	0.1	0.2

L2-Biases

0.1
0.4

L3-Weights

0.5	0.1
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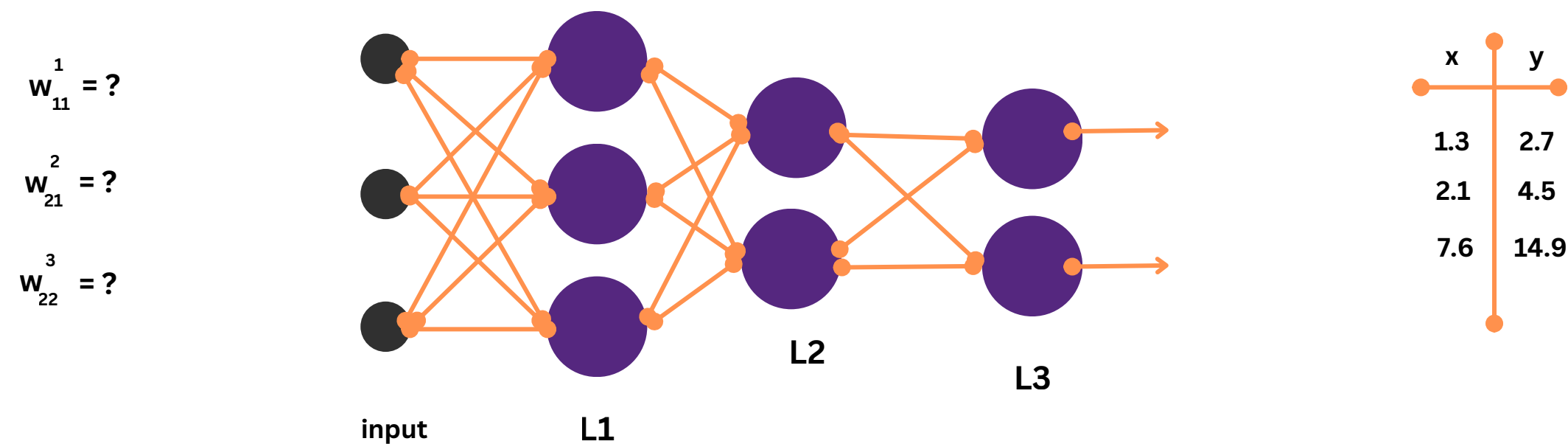
L3-Biases

0.5

Activation Functions:

L1: Relu
L2: Tanh
L3: Sigmoid

4. For the below feed-forward neural network architecture obtain the following weights after 1st Back Propagation iteration (epoch): [ignore the biases change during backpropagation]



inputs	L1-Weights			L1-Biases	L2-Weights			L2-Biases	L3-Weights		L3-Biases
0.3 0.5 0.8	0.1	0.2	0.2	0.2 0.2 0.1	0.1	0.2	0.2	0.1 0.4	0.5	0.1	0.5 0.3
	0.4	0.1	0.2		0.4	0.1	0.2		0.5	0.5	
	0.2	0.3	0.1								

● Activation Functions:

L1: Relu

L2: Tanh

L3: Sigmoid

● Loss Function: MSE

● Optimizer: GD

Initial weight value(weight have been initialized randomly) : 1.5

Initial bias value (weight have been initialized randomly) : 0.5

learning rate (alpha) = 0.1